# Lab Report: Training dynamics of deep models

## 1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 12 GB of memory. Each experiment was repeated 5 times with different random seeds.

## 2 Results

Figure 1 shows the f1 score during training.

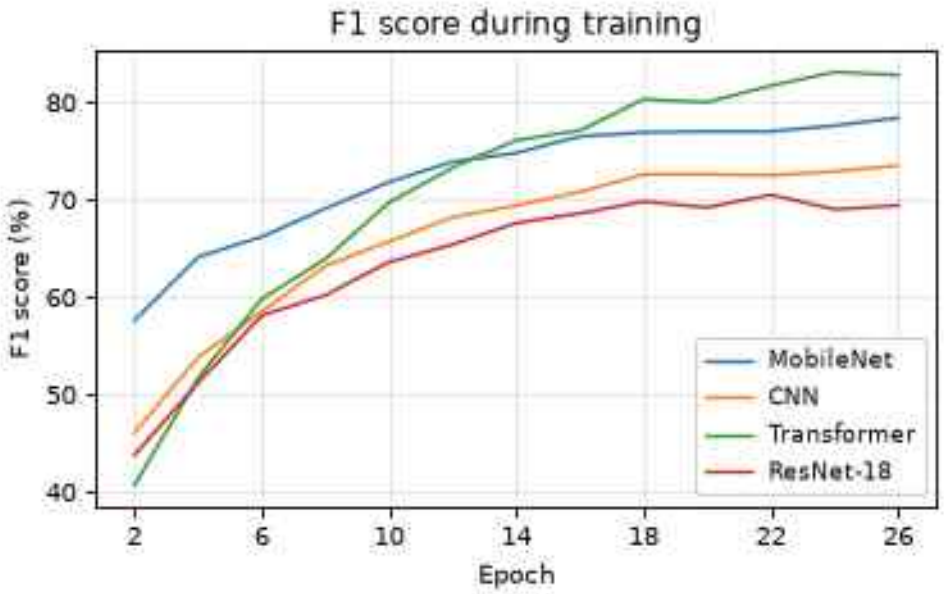



Figure 1: F1 score during training for the models in this experiment.

As shown in Figure 1, from epoch 20 to epoch 24, the F1 score of Transformer improves by a relative 2.7%. CNN reaches 67.3% F1 score at epoch 12. As shown in Figure 1, at epoch 2, CNN is 14.7 points better than Transformer.

Figure 2 shows the test accuracy by dataset.

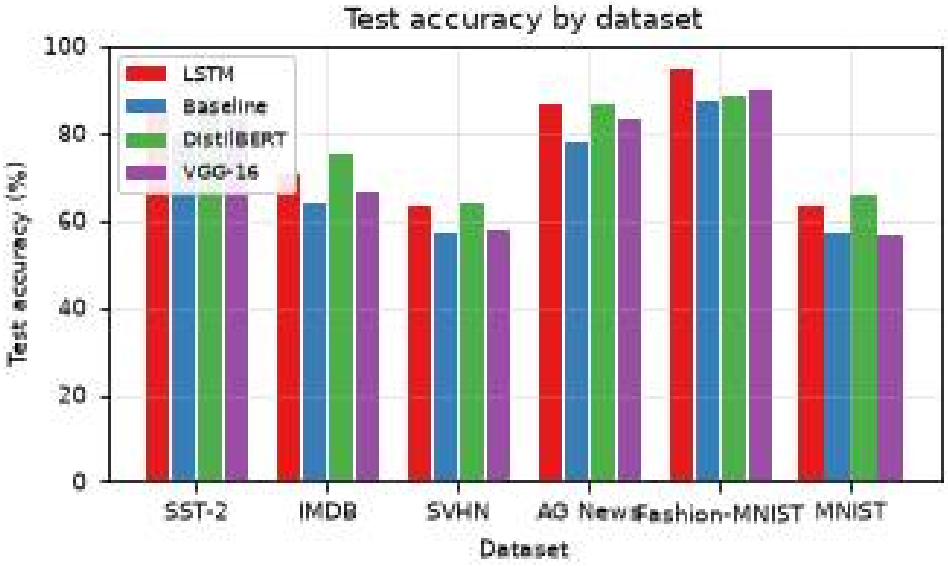



Figure 2: Test accuracy by dataset for the models in this experiment.

LSTM achieves 86.2% test accuracy on AG News (Figure 2). As shown in Figure 2, LSTM achieves over 85% test accuracy on AG News. On SST-2, DistilBERT is 55.5 points better than VGG-16. DistilBERT outperforms LSTM by 2.4 percentage points on MNIST (Figure 2).

## 3 Discussion

On CIFAR-10, SVM obtains an accuracy of 70.2%. We note that the training was stable for all models.